

Analysis First-Year Exam, January 2022, Part 2

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. For each integer $n \geq 1$, let $\chi_n = n\mathbf{1}_{[0,1/n]}$ be n times the indicator function of the interval $[0, 1/n]$. Prove that if $f : [0, 1] \rightarrow \mathbb{R}$ is continuous, then

$$\lim_{n \rightarrow \infty} \int_0^1 \chi_n(t) f(t) dt = f(0).$$

2. Let $a > 0$. When n is a positive integer and $t \geq 0$, write

$$f_n(t) = \frac{\sin nt}{1 + nt}.$$

Is the sequence $(f_n : n > 0)$ uniformly convergent on $[a, \infty)$? On $[0, \infty)$?

3. Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous. Prove that if

$$\int_0^1 f(t) t^{n+\frac{1}{2}} dt = 0$$

for each positive integer n , then f vanishes identically.

4. When $(f_n : n > 0)$ is a sequence of measurable real-valued functions on some measurable space, show that the following sets are measurable:

- (i) $\{\omega : f_n(\omega) \text{ alternates in sign}\}$
- (ii) $\{\omega : f_n(\omega) \text{ is eventually rational}\}$
- (iii) $\{\omega : \sum_{n>0} f_n(\omega) \text{ is absolutely convergent}\}.$

5. State the Monotone Convergence Theorem and the Fatou Lemma, and deduce one of these from the other.